

Filippo Ficarra

 FilippoFicarra |  Filippo Ficarra |  filippoficarra.github.io
 filippoficarra.careers@gmail.com |  +41762315690 |  Zürich

WORK EXPERIENCE

IBM Research Zürich – *Research Scientist Intern*

Jan 2025 - Jul 2025

- **Structure Elucidation from Mixture IR Spectra** |  | 
 - Trained, in collaboration with chemists, a **multimodal Transformer** to generate the structure of the target molecules from *mixture IR spectra* and the molecular formula of the target compound.
 - Aligned encoder representations with target molecule spectra to boost accuracy, achieving **~34% Top-1 Accuracy** (SOTA, outperforming human experts).
- **IBM Granite for Single Atom Catalysts Synthesis** | *DeepSpeed ZeRO-3, LangChain*
 - Built a **knowledge extraction** system for SAC literature using **LangChain**. Instruction-tuned **IBM Granite 3.2 8B** to enhance synthesis predictions with multitask and n-shot RAG setups.
- **Large Scale LLM Training** | *DeepSpeed ZeRO-3*
 - Trained **8 to 405 B** LLMs in a Slurm cluster on **2-2048 Nvidia GH200 GPUs**.

ETH Zürich – *Teaching Assistant*

Sep 2023 - Sep 2024

- Supervised students during Lab sessions. Tested assignments in advance and grade exam for **Large Language Models** and **Information Security Lab**. Material: LLMs  and InfoSec Lab .

ORBYTA TECH – *Backend Engineer Intern*

Mar 2022 - Jun 2022

- Designed and optimized a **REST API** backend for a B2B clothing platform, using C# with the ABP Framework and **MySQL** to deliver scalable and efficient data services.

JEToP (Student Association) – *Mobile Developer*

Nov 2021 - Aug 2022

- Developed and enhanced the **front-end** of a podcast application, implementing **multimedia playback**, dynamic routing, and interactive pages to improve user experience.

EDUCATION

ETH Zürich – M.Sc. in Computer Science

Sep 2022 - Nov 2024

- Major in **Machine Intelligence**. Relevant Courses: NLP, **Large Language Models**, **Computer Vision**, Probabilistic AI.
- Minor in **Information Security**. Relevant Courses: **Applied Cryptography**, Information Security Lab, Security of Wireless Networks, Network Security.
- GPA: **5.3/6.0**.

Politecnico di Torino – B.Eng. in Computer Engineering

Sep 2019 - Jul 2022

- Fully funded by **Merit-based Scholarship** “Percorso Giovani Talenti”. Relevant courses: Data Structures and Algorithms, Object-Oriented Programming.
- GPA: **110/110 cum laude**.

RESEARCH

ETH Zürich

Apr 2024 - Oct 2024

- **Master Thesis - Rycolab** | Supervisors: Ryan Cotterell, Alex Warstadt |  | 
 - **Title:** A Distributional Perspective on Word Learning in Neural Language Models.
 - Investigated how **LLMs** acquire words compared to **humans** by developing **surprisal-based metrics** and **training GPT-2** models from scratch on varying text amounts and linguistic sources; findings show that LLMs learn differently from humans, relying more on word frequency. This work was accepted as a long paper at **NAACL 2025**.

- **Semester Project - Rycolab | Supervisors: Ryan Cotterell, Alex Warstdat**
 - **Title:** How Multimodality Influences Word Acquisition: Do Images Help Language Models Mimic Human Behavior?
 - Analyzed how varying vision-to-text ratios influences language learning in **FLAVA multimodal models**, finding that higher visual input negatively impacts word learning efficiency.

- **Swiss AI Initiative – Rycolab, SwissAI, CSCS cluster**
 - Contributed to the training data and model architecture team for a large-scale academic **multimodal Vision-Video-LM**.

SELECTED PROJECTS

Lecture Transcriber [In progress]

- Building a **full-stack web application** that transcribes lectures using **OpenAI Whisper**; implemented **user authentication**, **lecture upload**, and a prototype **text editor**, with planned features such as structured lecture notes, slides, flashcards, and collaborative editing. Tech stack: **Next.js** (frontend), **FastAPI** (backend), **PostgreSQL** (database).

Sentiment Analysis on Tweets | | *TPUs, Google Cloud*

- Developed an ensemble model for sentiment analysis achieving **92.62% accuracy** on POS/NEG classification. Integrated **fine-tuned RoBERTa** with **sparsemax** attention, **BERTweet**, and **CLIP text features** to capture nuanced sentiment and enhance performance.

Probabilistic AI |

- Developed and applied advanced probabilistic AI and reinforcement learning models, including **Gaussian Processes regression** to predict pollution concentrations across city areas, SWA-Gaussian for improved predictive stability through stochastic weight averaging, and **Actor-Critic RL** agent to train a pendulum swing.

Computer Vision |

- Developed computer vision algorithms to perform feature extraction using Harris corner detection, **unsupervised image segmentation** with the Mean-Shift Algorithm, and object tracking using CONDENSATION. Implemented **Structure from Motion** to reconstruct 3D scenes from 2D images.

PUBLICATIONS

Ficarra, Filippo et al. (Apr. 2025). “A Distributional Perspective on Word Learning in Neural Language Models”. In: *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*. Albuquerque, New Mexico: Association for Computational Linguistics, pp. 11184–11207. DOI: [10.18653/v1/2025.naacl-long.558](https://doi.org/10.18653/v1/2025.naacl-long.558).

Ficarra, Filippo et al. (2025). “Language Model Enabled Structure Prediction from Infrared Spectra of Mixtures”. In: *AI for Accelerated Materials Design - NeurIPS 2025*. URL: <https://openreview.net/forum?id=3pAVbjWMXW>.

SKILLS

Programming Languages	Python , JavaScript, C#, Java, C, SQL, Bash
Python Ecosystem	NumPy, Pandas , Matplotlib, SciPy, Jupyter Notebook
ML Tools	PyTorch , Deepspeed , Hugging Face, LangChain , FastAPI, Streamlit
Technologies	Docker , Git , Slurm , IBM Blue Vela, IBM Cloud, Linux, LaTeX, MySQL